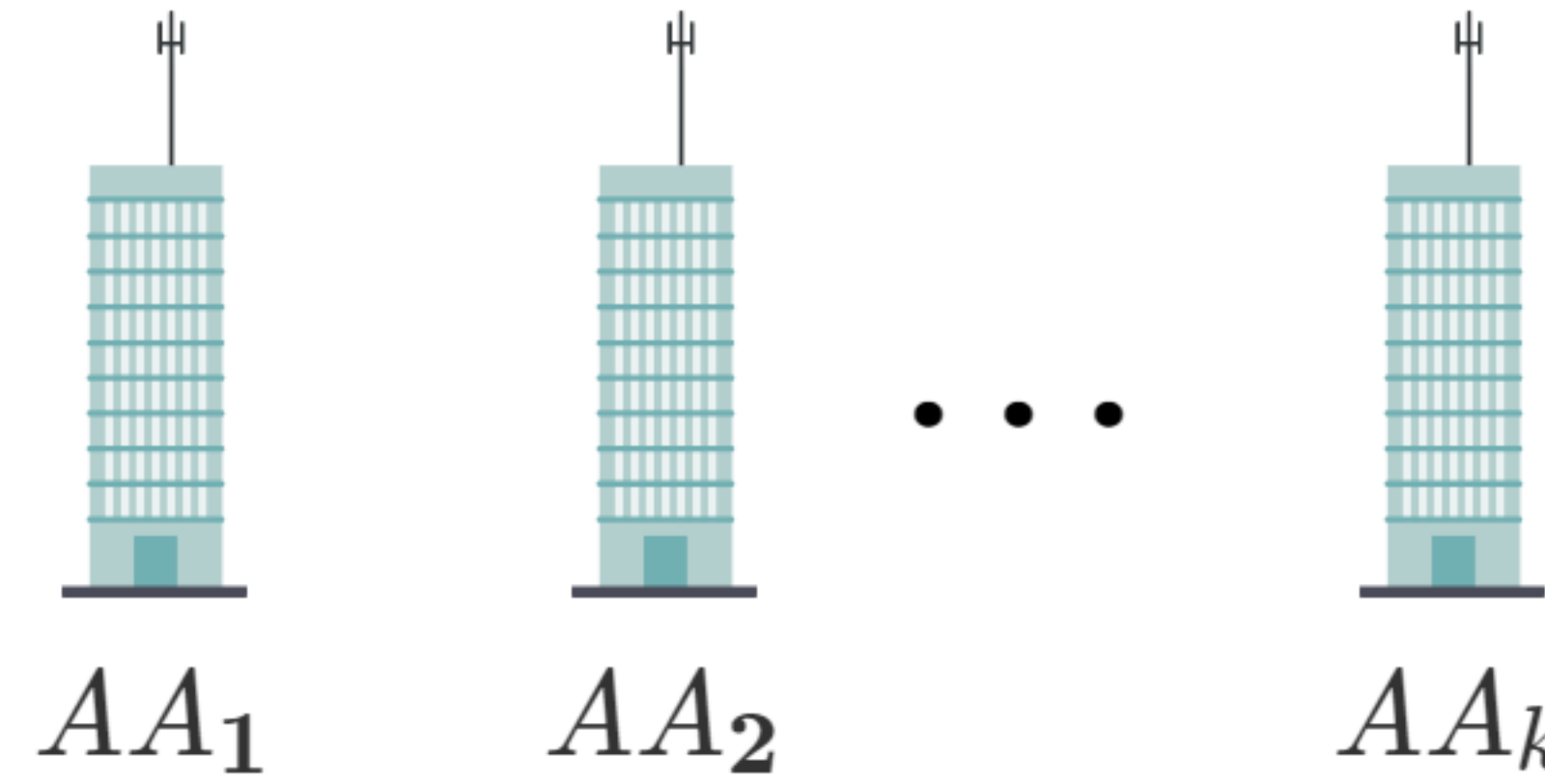




Attribute Authorities ($AA_1, AA_2, \dots, AA_k$)



Public Communication

Secure Channel

1.Setup:CA runs TrapGen and publishes mpk

2.Publish attribute public keys: Each AA uploads apk_i

3.Secret key distribution:CA /AA securely deliver attribute keys sk_x to the Signer

4.Policy announcement:Verifier specifies access policy f



Signer

6.Signature submission : Signer sends Message M



Verifier

5.Signature generation :

- 1)Compute $B_f = PubEval(apk, f) = AR_f + f(x)G$
- 2)Compute $R_f = TrapEval(sk_x, f)$
- 3) $e = SampleRight(A, G, R_f, T_G, u, \tau)$
- 4) $\sigma = (e, M, f)$

7.Verification :

- 1)Compute $B_f = PubEval(apk, f)$
- 2)Check $\|e\|_\infty < \beta$ and $[A \mid B_f] \cdot e = u$